

Exercise Sheet

Task 1 – Replacement Strategies

In an associative cache with four cache lines, the following sequence is executed: 0xB43, 0x427, 0x67C, 0xC6A, 0xB43, 0x406, 0xDA7, 0xE59, 0x406, 0xB43, 0x5E4, 0x427, 0xE59, 0xC6A, 0x406, 0xE59. Decide the cache's content for each of the following replacement strategies.

a) FIFO

Cells whose value does not change at that moment are intentionally left blank for better readability.

Line 1	Line 2	Line 3	Line 4
0xB43			
	0x427		
		0x67C	
			0xC6A
0x406			
	0xDA7		
		0xE59	
			0xB43
0x5E4			
	0x427		
		0xC6A	
			0x406
0xE59			

Total: 13 Misses

b) LFU with LRU Tie-Break

Cells whose value does not change at that moment are intentionally left blank for better readability.

Line 1	Line 2	Line 3	Line 4
0xB43			
	0x427		
		0x67C	
			0xC6A
	0x406		
		0xDA7	
			0xE59
		0x5E4	
			0x427
		0xE59	
			0xC6A

Total: 11 Misses

c) Clairvoyant

If multiple addresses won't appear again in the future, choose the left-most line to replace its content

Cells whose value does not change at that moment are intentionally left blank for better readability.

Line 1	Line 2	Line 3	Line 4
0xB43			
	0x427		
		0x67C	
			0xC6A
		0x406	
			0xDA7
			0xE59
0x5E4			
0xC6A			

Total: 9 Misses

Task 2 – Set-Associative Caches

Access sequence: 0xABC, 0xB43, 0xC6A, 0xABC 0xDD4, 0xC6A 0xE2F, 0xABC

Execute the above sequence in a set-associative cache with 2 cache sets (Set 0 and Set 1) and 4 cache lines altogether and uses LRU as replacement policy. An address is put into a cache set that has the same number as its LSB. Decide for each access whether a cache hit or a cache miss happens. If a cache miss occurs, elaborate what type of miss happened.

Cells whose value does not change at that moment are intentionally left blank for better readability.

Set 0		Set 1		
Zeile 1	Zeile 2	Zeile 3	Zeile 4	Hit/Miss/Typ
0xABC				Compulsory
		0xB43		Compulsory
	0xC6A			Compulsory
				Hit
	0xDD4			Compulsory
0xC6A				Conflict
			0xE2F	Compulsory
	0xABC			Capacity